

Week 1: Introduction (II)

PLSC 309 Quantitative Political Analysis

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What Did We Learn Last Time?

- What do political scientists do?
- Normative vs. Positive Research
- Types of Positive Research
 - Theoretical Research
 - Empirical Research
 - Quantitative Research
 - Qualitative Research

Today's Agenda

- ① How Do Political Scientists Conduct Research?
 - Asking a research question
 - Developing a theory and a hypotheses
 - Evaluating hypotheses with empirical evidence
- ② Lab Session
 - Installing R and RStudio
 - Introduction to R

Part 1. How Do Political Scientists Conduct Research?

What Is Research?

What Is Research?

- It is “the systematic investigation into and study of materials and sources in order to establish facts and reach new conclusions.” (Oxford Languages Dictionary)
- Doing research means transitioning from being a **consumer of knowledge** to becoming both a **consumer and producer of knowledge**.
- “If we knew what we were doing, it would not be called research.’’ (Albert Einstein)

Infographic Explains the Path of a (Ph.D) Research

Imagine a circle that contains all of human knowledge:



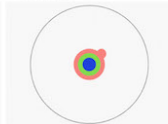
By the time you finish elementary school, you know a little:



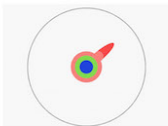
By the time you finish high school, you know a bit more:



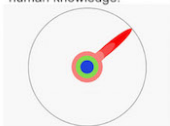
With a bachelor's degree, you gain a specialty:



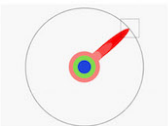
A master's degree deepens that specialty:



Reading research papers takes you to the edge of human knowledge:



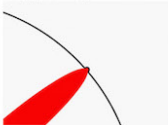
Once you're at the boundary, you focus:



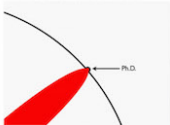
You push at the boundary for a few years:



Until one day, the boundary gives way:



And, that dent you've made is called a Ph.D.:



Of course, the world looks different to you now:



So, don't forget the bigger picture:



Keep pushing.

Step One: Asking a Research Question

Research Questions: Where Research Begins

What is a Research Question?

A **research question** is the central question that a study is designed to answer.

More specifically, it is:

- A problem that we do not know or fully understand yet (A puzzle).
- A question with a well-defined and researchable answer.
- A question that can be answered convincingly through a systematic research design.

Some Classic Research Questions in Political Science

American Politics

- How do voters form political preferences and make electoral choices?
- Do elected officials respond to public opinion?
- How are political interests organized and represented?

Comparative Politics

- Why do some countries become democratic while others do not?
- How do authoritarian regimes maintain power and legitimacy?

International Relations

- Why do states go to war?
- How is international cooperation possible?

Distinction between Research Topic and Research Question

Research Topic:

- a **subject or issue** that a researcher is interested in when conducting research.
 - American political institutions (Congressional politics, bureaucratic politics, judicial politics, state politics, local politics, interest group politics. . .)
 - International conflict, alliances, diplomacy, foreign policy. . .
 - International trade, foreign direct investment, immigration, . . .

A research topic is a broad area of interest, while a research question is a specific question you want to answer within that area.

Where Do Research Ideas Come From?

1 Observations

- Personal experiences
- Current events and news
- Historical events
- The world around us



Newton's observation of a falling apple

Where Do Research Ideas Come From? (Cont.)

② Existing Research

- Identifying important but unanswered questions
- Identifying new scope conditions¹ for existing theories
- Reconciling conflicting findings
- Proposing alternative explanations by challenging assumptions in existing theories
- Extending theories to different levels of analysis (e.g., from states/institutions to individuals).

¹Scope conditions specify the circumstances under which a theory is expected to hold—for example, particular countries, historical periods, political institutions, or populations.

What Makes a Compelling Research Question?

A compelling research question is one that is:

- **Interesting** – Sparks curiosity not only for you, but also for the broader research community and even the society, and has the potential to inspire further research.
- **Important** – Answering the question improves our understanding of a related phenomenon and advances knowledge in the field.

A good research question must be both interesting and important.

Step Two: Developing a Theory and a Hypothesis

What Is A Theory?

- A **theory** is a **logical explanation** for why the answer to your research question might be what it is.
- Rather than simply describing what happened.

Theory Is A Causal Story

A theory tells a causal story:

- What causes the outcome?
- Why does the cause produce the outcome?
- Under what conditions does the relationship hold? (*if the theory specifies scope conditions*)

In other words,

X causes Y because of M (under conditions C)

where

- **X** = Independent variable (IV, the cause)
- **Y** = Dependent variable (DV, the outcome)
- **M** = Mediator / Causal mechanism (why X affects Y)
- **C** = Moderator (when X affects Y)

From Theory to Hypothesis

A **theory** is a (general) **causal explanation** of your research question.

A **hypothesis** is a **theory-based, testable prediction** about what we expect to observe if the theory is correct.

What Makes a Good Hypothesis?

A good hypothesis should be:

- **Theory-driven** – Derived from a logical causal explanation.
- **Specific** – Clearly states the expected relationship between IV and DV.
- **Testable** – Can be evaluated using empirical evidence.
- **Falsifiable** – Can be supported or contradicted by the evidence.

Example: Economic Voting Theories

Political behavior: What explains how people vote?

- In political science, economic voting is a theoretical perspective which argues that voter behavior is heavily influenced by the economic conditions in their country at the time of the election.

Explanation 1: Sociotropic Voting Theory

Research Question

What explains how people vote?

Theory

Voters evaluate the government's management of the national economy. When the economy performs well, voters reward incumbents; when it performs poorly, voters punish incumbents.

Hypothesis

Higher national economic growth increases the probability that the incumbent party wins the election.

Explanation 2: Pocketbook Voting Theory

Research Question

What explains how people vote?

Theory

Voters evaluate how government policies affect their own economic well-being. Individuals who are financially better off are more likely to support the incumbent.

Hypothesis

Individuals whose personal financial situation has improved are more likely to vote for the incumbent party.

Explanation 3: Prospective Voting Theory

Research Question

What explains how people vote?

Theory

Voters support the candidate or party they expect will deliver better economic performance in the future.

Hypothesis

Individuals who expect the economy to improve under the incumbent are more likely to vote for the incumbent party.

Competing Economic Voting Theories and Hypotheses

Theory	Voters Base Their Decisions On...	Hypothesis
Sociotropic Voting	National economic performance	Better national economic performance increases support for the incumbent.
Pocketbook Voting	Personal financial situation	Individuals whose personal financial situation improves are more likely to support the incumbent.
Prospective Voting	Expectations about future economic performance	Individuals who expect the economy to improve under the incumbent are more likely to support the incumbent.

The same research question can have multiple plausible explanations, leading to competing theories and testable hypotheses.

In-Class Exercise: Identify the Variables

For each hypothesis, identify the **independent variable (IV)** and **dependent variable (DV)**.

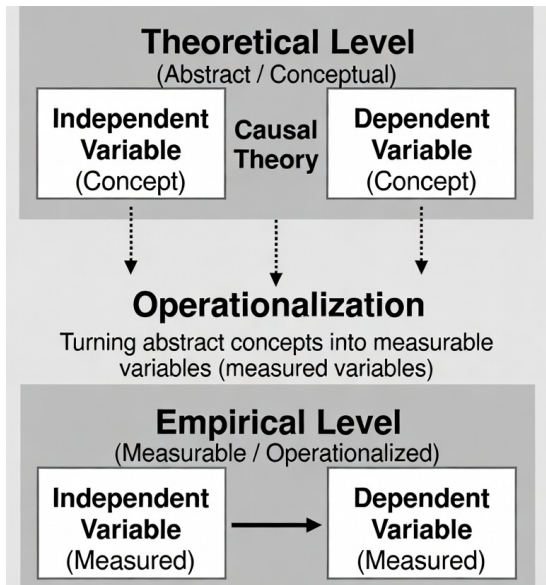
- ① People with higher levels of education are more likely to vote.
- ② Higher unemployment reduces support for the incumbent party.
- ③ Democracies are less likely to go to war with one another than nondemocracies.
- ④ Support for democratic institutions is higher among individuals with greater political knowledge.

In-Class Exercise: Answers

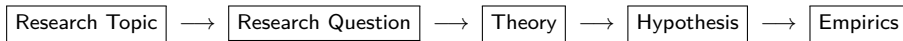
Hypothesis	Independent Variable (IV)	Dependent Variable (DV)
1	Education	Voting
2	Unemployment	Incumbent support
3	Regime type	War
4	Political knowledge	Support for democratic institutions

Step Three: Evaluating Hypotheses with Quantitative Evidence

From Theory to Empirics



The Research Process (Roadmap)



Research Topic

What broad area are you interested in?

Research Question

What specific question do you want to answer?

Theory

What explains the outcome?

Cause → Mechanism → Outcome (+ Scope Conditions)

Hypothesis

What observable relationship does the theory predict?

Empirics

What evidence can we use to evaluate the hypothesis?

Empirical Evidence This is where we are headed: the rest of this course focuses on how to provide credible quantitative evidence to evaluate hypotheses.

Key Takeaways

After today's lecture, you should be able to:

- Explain what makes a compelling research question.
- Define a theory and a hypothesis.
- Identify the independent and dependent variables in a hypothesis.
- Understand the basic research process in political science.

What Comes Next?

Next Lecture (September 1):

- From Concepts to Data

The rest of the semester:

- Estimating relationships between independent and dependent variables using observational data
- Moving from correlation to causality

Part 2. Getting Started with R and RStudio

What Is R?

R is a **free, open-source programming language** designed for:

- Data analysis
- Statistical computing
- Data visualization
- Reproducible research

Why Do We Use R?

- **Free** – No license required.
- **Widely used** - Popular software in academia, government, and industry.
- **Reproducible** – Every analysis can be reproduced from code.
- **Flexible** – Supports data management, visualization, and statistical modeling.
- **Extensible** – Thousands of user-developed packages are freely available.

What Is RStudio?

RStudio is an Integrated Development Environment (IDE) for R.

Think of it as your workspace for writing, running, and organizing R code.

R is the engine.

RStudio is the dashboard.

Today's Lab

- ① Install R
- ② Install RStudio
- ③ Explore the RStudio interface
- ④ Run your first R commands

Step 1: Install R

R is the programming language that we will use throughout this course.

Download the latest version of R from:

<https://cran.r-project.org>

Choose the installer for your operating system:

- Windows
- macOS
- Linux

Install using the default settings.

Step 2: Install RStudio

RStudio is an integrated development environment (IDE) for R.

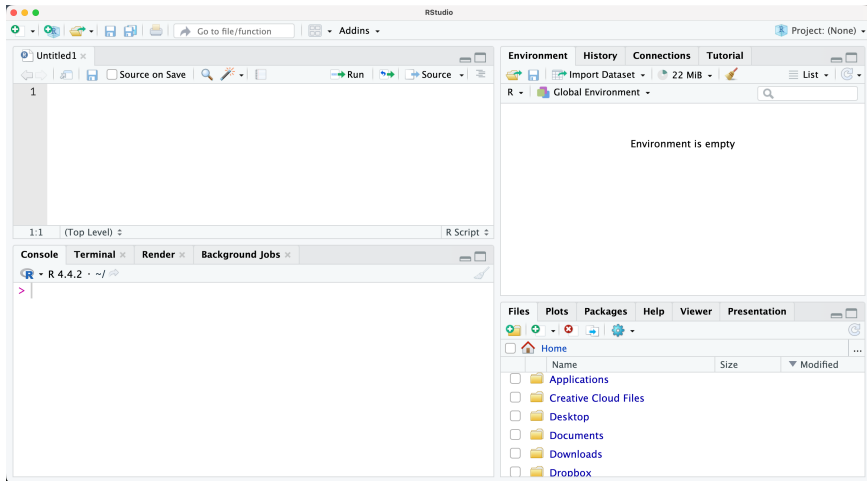
Download RStudio Desktop from:

<https://posit.co/download/rstudio-desktop/>

Install using the default settings.

The RStudio Interface

When you open RStudio, you will see four main panes.



The RStudio Interface

Source (Top Left)

- Write and save R scripts
- Edit code before running

Environment / History (Top Right)

- View objects currently stored in memory
- Review previously executed commands

Console (Bottom Left)

- Execute R commands
- View output and error messages

Files / Plots / Packages / Help (Bottom Right)

- Browse files
- View plots
- Manage packages
- Access documentation

Your First R Commands

R can be used as a calculator.

```
2 + 3
```

```
## [1] 5
```

```
10 - 4
```

```
## [1] 6
```

```
5 * 8
```

```
## [1] 40
```

```
20 / 4
```

```
## [1] 5
```

```
2^5
```

```
## [1] 32
```


Mathematical Functions

R includes many built-in functions.

```
sqrt(36)
```

```
## [1] 6
```

```
abs(-15)
```

```
## [1] 15
```

```
log(100)
```

```
## [1] 4.60517
```

```
exp(1)
```

```
## [1] 2.718282
```

```
round(pi, digits=2)
```

```
## [1] 3.14
```

Creating Objects

Use `<-` to assign values to objects.

```
x <- 10
```

```
y <- 25
```

```
x
```

```
## [1] 10
```

```
y
```

```
## [1] 25
```

Working with Objects

Once an object is created, it can be reused, overwritten, and updated.

```
x <- 10  
y <- 25  
x + y
```

```
## [1] 35
```

```
y - x
```

```
## [1] 15
```

```
x <- 20  
x + y
```

```
## [1] 45
```

```
y - x
```

```
## [1] 5
```

Adding Comments

Use # to write notes.

```
# This is a comment
```

```
x <- 10
```

```
# Calculate x squared
```

```
x^2
```

```
## [1] 100
```

Coding Exercise: Creating and Using Objects

Create the following objects with the assignment operator: `<-`:

- `votes_a` equal to 1250
- `votes_b` equal to 980

Then use those objects to calculate:

- 1 The total number of votes
- 2 The difference in votes
- 3 Candidate A's share of the total vote

Coding Exercise: Answers

```
votes_a <- 1250

votes_b <- 980

total_votes <- votes_a + votes_b

vote_difference <- votes_a - votes_b

vote_share_a <- votes_a / total_votes

total_votes

## [1] 2230

vote_difference

## [1] 270

vote_share_a

## [1] 0.5605381
```

- **September 1 (Tue):** From Concepts to Data
- **September 2 (Wed):** No office hours due to conference travel. Please email me if you have any questions.
- **September 3 (Thu):** No class (Instructor traveling to APSA Conference).